Financial Risk Manager (FRM®) Examination

2013 Practice Exam

PART II

Answer Sheet

	a.	b.	c.	d.
1.	○	○	○	○
2.	○	○	○	○
3.	○	○	○	○
4.	○	○	○	○
5.	○	○	○	○
6.	○	○	○	○
7.	○	○	○	○
8.	○	○	○	○
9.	○	○	○	○
10.	○	○	○	○
11.	○	○	○	○
12.	○	○	○	○
13.	○	○	○	○

	a.	b.	c.	d.
14.	○	○	○	○
15.	○	○	○	○
16.	○	○	○	○
17.	○	○	○	○
18.	○	○	○	○
19.	○	○	○	○
20.	○	○	○	○

Correct way to complete

1.	●	●	●	●

Wrong way to complete

1.	⊘	⊖	✓	⊗

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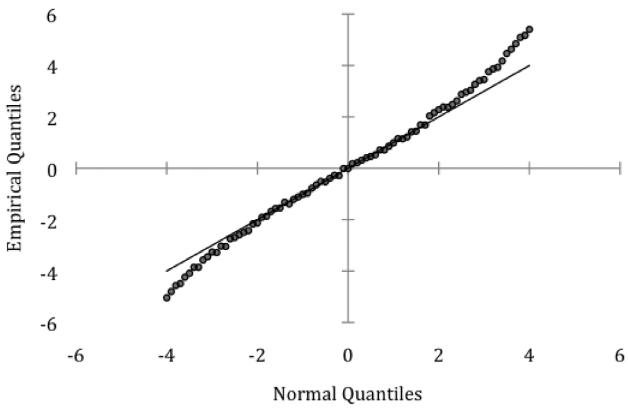
2013 Practice Exam

PART II

Questions

1. You are examining a sample of return data. As a first step, you construct a QQ plot of the data as shown below:



Based on an examination of the QQ plot, which of the following statements is correct?

a. The returns are normally distributed.
b. The return distribution has thin tails relative to the normal distribution.
c. The return distribution is negatively skewed relative to the normal distribution.
d. The return distribution has fat tails relative to the normal distribution.

2. The annual mean and volatility of a portfolio are 10% and 40%, respectively. The current value of the portfolio is GBP 1,000,000. How does the 1-year 95% VaR that is calculated using a normal distribution assumption (normal VaR) compare with the 1-year 95% VaR that is calculated using the lognormal distribution assumption (lognormal VaR)?

a. Lognormal VaR is greater than normal VaR by GBP 13,040
b. Lognormal VaR is greater than normal VaR by GBP 17,590
c. Lognormal VaR is less than normal VaR by GBP 13,040
d. Lognormal VaR is less than normal VaR by GBP 17,590

3. Bennett Bank extends a 5% APR (annual percentage rate) USD 100,000 30-year mortgage requiring monthly payments. If the mortgage is structured so that it requires interest-only payments for the first 5 years, after which point it becomes a self-amortizing mortgage, what would be the portion of the monthly payment applied to the principal in the 61st month?

a. USD 167.92
b. USD 174.60
c. USD 584.59
d. USD 591.27

4. Let X be a random variable representing the daily loss of your portfolio. The "peaks over threshold" (POT) approach considers a threshold value, u, of X and the distribution of excess losses over this threshold. Which of the following statements about this application of extreme value theory is correct?

- **a.** To apply the POT approach, the distribution of X must be elliptical and known.
- **b.** If X is normally distributed, the distribution of excess losses requires the estimation of only one parameter, β, which is a positive scale parameter.
- **c.** To apply the POT approach, one must choose a threshold, *u*, which is high enough that the number of observations in excess of *u* is zero.
- **d.** As the threshold, *u*, increases, the distribution of excess losses over *u* converges to a generalized Pareto distribution.

5. A risk analyst is comparing the use of parametric and non-parametric approaches for calculating VaR and is concerned about some of the characteristics present in the loss data. Which of the following distribution characteristics would make parametric approaches the favored method to use?

- **a.** Skewness in the distribution
- **b.** Fat tails in the distribution
- **c.** Scarcity of high magnitude loss events
- **d.** Heteroskedasticity in the distribution

6. Lin Ping is valuing a 1-year credit default swap (CDS) contract which will pay the buyer 75% of the face value of a bond issued by Xiao Corp. immediately after a default by Xiao. To purchase this CDS, the buyer will pay the CDS spread, which is a percentage of the face value, once at the end of the year. Lin estimates that the risk-neutral default probability for Xiao is 5% per year. The risk-free rate is 3% per year. Assuming defaults can only occur halfway through the year and that the accrued premium is paid immediately after a default, what is the estimate for the CDS spread?

- **a.** 380 basis points
- **b.** 385 basis points
- **c.** 390 basis points
- **d.** 400 basis points

7. You are the risk manager at Vision, a small fixed-income hedge fund that specializes in bank debt. Vision's strategy utilizes both relative value and long-only trades using credit default swaps (CDS) and bonds. One of the new traders has the positions described in the table below.

Bank	Position	Credit Rating
SBU	Long USD 10 million CDS	A
Stanos	Long USD 5 million bond	BB+
CAB	Short USD 10 million CDS	A

Some of Vision's newest clients are restricted from withdrawing their funds for three years. You are currently evaluating the impact of various default scenarios to estimate future asset liquidity. You have estimated that the marginal probability of default of the Stanos bond is 5% in Year 1, 10% in Year 2, and 15% in Year 3. What is the probability that the bond makes coupon payments for 3 years and then defaults at the end of Year 3?

- **a.** 13%
- **b.** 15%
- **c.** 27%
- **d.** 73%

8. Consider a 1-year maturity zero-coupon bond with a face value of USD 1,000,000 and a 0% recovery rate issued by Company A. The bond is currently trading at 80% of face value. Assuming the excess spread only captures credit risk and that the risk-free rate is 5% per annum, the risk-neutral 1-year probability of default on Company A is closest to which of the following?

- **a.** 2%
- **b.** 14%
- **c.** 16%
- **d.** 20%

9. Portland General Electric (PGE) was an Enron subsidiary that was able to survive after the Enron implosion. At that time, there was a trend towards electric utility downgrades, particularly for those utilities operating within larger corporate structures. PGE survived in part due to ring-fencing. Which of the following statements about ring fencing is correct?

- **a.** A ring-fencing assets approach is typically only useful when a low quality firm wants to finance a high-quality project.
- **b.** When ring-fencing assets, options for credit enhancement include overcollateralization and financial guarantees provided by the parent against default of the subsidiary.
- **c.** A subsidiary holding the ring-fenced assets may be able to gain a higher credit rating than the parent, allowing it to issue bonds on the assets at a lower cost.
- **d.** Because the parent does not retain an equity interest in the subsidiary holding the ring-fenced assets, the subsidiary is not consolidated on the parent's balance sheet.

10. Bank A, a large international bank, engages in trading with counterparties throughout the world. Recently, it has started to pay more attention to wrong-way risk in its trading book. Which one of the following four scenarios would serve as an example of wrong-way risk from Bank A's perspective?

- **a.** Bank A has a large exposure to Bank B's equity, and Bank B offers to sell put options with long maturities on its own equity to Bank A.
- **b.** Bank A enters into a medium-term repurchase agreement with Bank B using several different types of debt issued by bank B as collateral.
- **c.** Bank A actively manages its credit portfolio using credit default swaps, and decides to sell long-term credit protection to Bank B.
- **d.** Bank A enters into a forward rate agreement with Bank B to deliver at LIBOR+2.5%.

11. The Chief Risk Officer of your bank has put you in charge of operational risk management. As a first step, you collect internal data to estimate the frequency and severity of operational-risk-related losses. The table below summarizes your findings:

Frequency Distribution			Severity Distribution	
Number of Occurrences	**Probability**		**Loss (USD)**	**Probability**
0	0.6		1,000	0.5
1	0.3		100,000	0.4
2	0.1		1,000,000	0.1

Based on this information, what is your estimate of the expected loss due to operational risk?

- **a.** USD 20,000
- **b.** USD 70,250
- **c.** USD 130,600
- **d.** USD 140,500

12. In its efforts to enhance its enterprise risk management function, Countryside Bank introduced a new decision-making process based on economic capital that involves assessing sources of risk across different business units and organizational levels. Which of the following statements regarding the correlations between these risks is correct?

- **a.** Correlations between the risks in the asset and liability sides of the balance sheet can be changed by management decisions.
- **b.** Generally, correlations between broad risk types such as credit, market, and operational risk are well understood and are easy to estimate at the individual firm level.
- **c.** Correlations between business units are only relevant in deciding total firm-wide economic capital levels and are not relevant for decisions at the individual business unit or project level.
- **d.** The introduction of correlations into firm-wide risk evaluation will result in a total VaR that, in general, is greater than or equal to the sum of individual business unit VaRs.

13. In recent years, large dealer banks financed significant fractions of their assets using short-term, often overnight, repurchase (repo) agreements in which creditors held bank securities as collateral against default losses. The table below shows the quarter-end financing of four broker-dealer banks. All values are in USD billions:

	Bank A	Bank B	Bank C	Bank D
Financial instruments owned	823	629	723	382
Pledged as collateral	272	289	380	155

In the event that repo creditors become nervous about a bank's solvency, which bank is least vulnerable to a liquidity crisis?

a. Bank A
b. Bank B
c. Bank C
d. Bank D

14. Galileo Vehicles (GV) and Leonardo Motors (LM) are both leading car manufacturers in hybrid car designs. Earlier this year, both companies introduced new hybrid models that are comparable to each other in almost every category. However, after both companies release pricing for their new models, LM's model is 20% less expensive than GV's. As a result, GV's stock price declined sharply while LM's stock price rose dramatically. Subsequently LM and GV announce that they have entered into merger discussions where the terms of the planned merger would give GV shareholders 1 share of LM per 3 shares of GV previously held. Post the announcement, GV's stock is trading at USD 20 and LM's stock is trading at USD 58. If you are confident that the merger will be completed, assuming zero transaction costs, which of the following investments should you make?

a. Buy 300 shares of GV and short 100 shares of LM.
b. Short 300 shares of GV and buy 100 shares of LM.
c. Buy 300 shares of GV and buy 100 shares of LM.
d. Short 300 shares of GV and short 100 shares of LM.

15. At the end of 2007, Chad & Co.'s pension had USD 350 million worth of assets that were fully invested in equities and USD 180 million in fixed-income liabilities with a modified duration of 14. In 2008, the widespread effects of the subprime crisis hit the pension fund, causing its investment in equities to lose 50% of their market value. In addition, the immediate response from the government — cutting interest rates — to salvage the situation, caused bond yields to decline by 2%. What was the change in the pension fund's surplus in 2008?

a. USD -55.4 million
b. USD -124.6 million
c. USD -225.4 million
d. USD -230.4 million

16. As investors found out that highly-rated securities backed by subprime mortgages were not risk-free, the subprime crisis affected other asset classes. Which of the following mechanisms played an important role in the transmission of the crisis from the subprime sector to other asset classes?

a. Impact of cross-default clauses linking industrial bonds and commercial mortgages held by banks
b. Increase in repo haircuts causing banks to sell off assets to meet collateral calls
c. Tightening of discount window lending standards by the U.S. Federal Reserve
d. Exercise of credit default swap contracts tied to subprime mortgage pools held in off-balance sheet vehicles

17. In the years leading up to the collapse of the Icelandic banking system, how did the relationship between the Central Bank of Iceland (CBI) and the Icelandic banks change?

a. The CBI supplemented credit ratings from the major rating agencies with market-implied ratings when determining the liquidity to provide to the banks.
b. The CBI began to issue loans to the banks that were denominated in Euros.
c. The CBI began to issue more loans to the banks that were collateralized by the bonds of other Icelandic banks.
d. The CBI began to issue loans to the banks that were denominated in U.S. dollars.

18. A portfolio has USD 2 million invested in Stock A and USD 1 million invested in Stock B. The 95% 1-day VaR for each individual position is USD 40,000. The correlation between the returns of Stock A and Stock B is 0.5. While rebalancing, the portfolio manager decides to sell USD 1 million of Stock A to buy USD 1 million of Stock B. Assuming that returns are normally distributed and that the rebalancing does not affect the volatility of the individual stocks, what effect will this have on the 95% 1-day portfolio VaR?

a. There will be no effect.
b. It will increase by USD 20,370.
c. It will increase by USD 21,370.
d. It will increase by USD 22,370.

19. In calculating its risk-adjusted return on capital, your bank uses a capital charge of 2.50% for revolving credit facilities with a loan equivalent factor of 0.35 assigned to the undrawn portion. Recently, you have become concerned that the protective covenants embedded in these loans are weak and may not prevent customers from drawing on the facilities during times of stress. As such, you have recommended doubling the loan equivalent factor to 0.70. This recommendation has met with resistance from the loan origination team, and senior management has asked you to quantify the impact of your recommendation. For a typical facility that has an original principal of USD 1 billion and is 30% drawn, how much additional economic capital would have to be allocated if you increase the loan equivalent factor from 0.35 to 0.70?

- **a.** USD 3.50 million
- **b.** USD 6.13 million
- **c.** USD 8.75 million
- **d.** USD 13.63 million

20. Which of the following statements regarding frictions in the securitization of subprime mortgages is correct?

- **a.** The arranger will typically have an information advantage over the originator with regard to the quality of the loans securitized.
- **b.** The originator will typically have an information advantage over the arranger, which can create an incentive for the originator to collaborate with the borrower in filing false loan applications.
- **c.** The major credit rating agencies are paid by investors for their rating service of mortgage-backed securities, and this creates a potential conflict of interest.
- **d.** The use of escrow accounts for insurance and tax payments eliminates the risk of foreclosure.

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Answers

	a.	b.	c.	d.
1.	○	○	○	●
2.	○	○	●	○
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4.	○	○	○	●
5.	○	○	●	○
6.	○	○	●	○
7.	●	○	○	○
8.	○	○	●	○
9.	○	○	●	○
10.	●	○	○	○
11.	○	●	○	○
12.	●	○	○	○
13.	●	○	○	○

	a.	b.	c.	d.
14.	○	●	○	○
15.	○	○	●	○
16.	○	●	○	○
17.	○	○	●	○
18.	○	○	○	●
19.	○	●	○	○
20.	○	●	○	○

Correct way to complete

1.	●	●	●	●

Wrong way to complete

1.	⊘	⊖	✓	⊗

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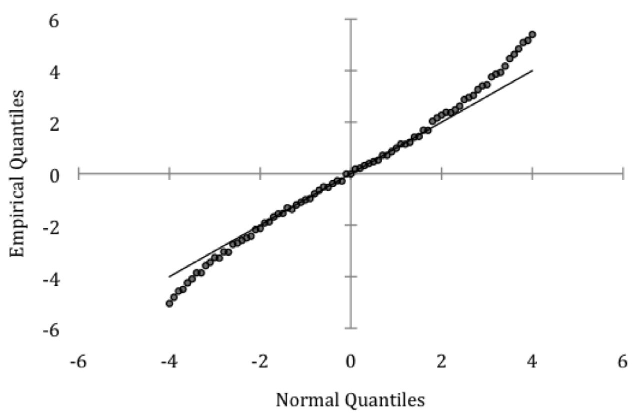
Explanations

1. You are examining a sample of return data. As a first step, you construct a QQ plot of the data as shown below:



Based on an examination of the QQ plot, which of the following statements is correct?

a. The returns are normally distributed.
b. The return distribution has thin tails relative to the normal distribution.
c. The return distribution is negatively skewed relative to the normal distribution.
d. The return distribution has fat tails relative to the normal distribution.

Correct answer: d

Explanation: This Q-Q plot has steeper slopes at the tails of the plot, which indicate fat tails in the distribution.

A normal distribution would result in a linear QQ plot. A distribution with thin tails would produce a QQ plot with less steep slopes at the tails of the plot than a linear relationship, while this one is steeper at the tails. It is not a negatively skewed distribution, as the Q-Q plot is symmetric.

Reference: Kevin Dowd, *Measuring Market Risk, 2nd Edition*, Chapter 3, pp. 75, 77.
AIM: Describe the use of QQ plots for identifying the distribution of data.
Section: Market Risk Management and Measurement

2. The annual mean and volatility of a portfolio are 10% and 40%, respectively. The current value of the portfolio is GBP 1,000,000. How does the 1-year 95% VaR that is calculated using a normal distribution assumption (normal VaR) compare with the 1-year 95% VaR that is calculated using the lognormal distribution assumption (lognormal VaR)?

- **a.** Lognormal VaR is greater than normal VaR by GBP 13,040
- **b.** Lognormal VaR is greater than normal VaR by GBP 17,590
- **c.** Lognormal VaR is less than normal VaR by GBP 13,040
- **d.** Lognormal VaR is less than normal VaR by GBP 17,590

Correct answer: c

Explanation: Normal VaR is calculated as follows:

Normal VaR = 0.1 – (1.645 * 0.4) = 0.558 (dropping negative sign)

and lognormal VaR is calculated as follows:

Lognormal VaR = 1 – exp [0.1 – (1.645 * 0.4)] = 0.4276

Hence, Lognormal VaR is smaller than Normal VaR by: 13.04% per year. With a portfolio of GBP 1,000,000 this translates to GBP 13,040.

Reference: Kevin Dowd, *Measuring Market Risk,* 2nd Edition, Chapter 3.
AIMS: Calculate VaR using a parametric estimation approach assuming that the return distribution is either normal or lognormal.
Section: Market Risk Management and Measurement

3. Bennett Bank extends a 5% APR (annual percentage rate) USD 100,000 30-year mortgage requiring monthly payments. If the mortgage is structured so that it requires interest-only payments for the first 5 years, after which point it becomes a self-amortizing mortgage, what would be the portion of the monthly payment applied to the principal in the 61st month?

- **a.** USD 167.92
- **b.** USD 174.60
- **c.** USD 584.59
- **d.** USD 591.27

Correct answer: a

Explanation: The principal payment for the 61st month is equal to the total monthly payment for the 61st month minus the total interest only payment for that month. First calculate the total monthly payment as shown below:

Total Monthly Payment = Mortgage payment factor * Principal balance

Mortgage payment factor: $r(1+r)^n / (1+r)^{n-1}$

where

r = the interest rate, and
n = the number of payments over the loan term.

Note that the interest rate corresponds to the frequency of the payment, so when using a monthly payment as in this example, the annual percentage rate (APR) must be divided by 12.

In this case, given that the monthly interest rate equals 0.0041667 (0.05 / 12) and 300 monthly payments will be made in the 25 remaining years of the loan, the mortgage payment factor is:

$[0.0041667 * (1.0041667)^{300}] / (1.0041667^{300} - 1) = .0058459.$

So the total monthly payment equals .0058459 * 100,000 or USD 584.59.

Next, compute the monthly interest payment, which is equal to 100,000 * (0.05 / 12) or USD 416.67.

Hence, the correct answer is 584.59 – 416.67 or 167.92, which reflects the principal portion of the 61st month's payment.

Reference: Frank Fabozzi, Anand Bhattacharya, and William Berliner, *Mortgage Backed Securities, 2nd Edition* (Hoboken: John Wiley & Sons, 2006), Chapter 1, p. 13.
AIMS: Calculate the mortgage payment factor. Understand the allocation of loan principal and interest over time for various loan types.
Section: Market Risk Management and Measurement

4. Let X be a random variable representing the daily loss of your portfolio. The “peaks over threshold” (POT) approach considers a threshold value, u, of X and the distribution of excess losses over this threshold. Which of the following statements about this application of extreme value theory is correct?

- **a.** To apply the POT approach, the distribution of X must be elliptical and known.
- **b.** If X is normally distributed, the distribution of excess losses requires the estimation of only one parameter, β, which is a positive scale parameter.
- **c.** To apply the POT approach, one must choose a threshold, *u*, which is high enough that the number of observations in excess of *u* is zero.
- **d.** As the threshold, *u*, increases, the distribution of excess losses over *u* converges to a generalized Pareto distribution.

Correct answer: d

Explanation: The distribution of excess losses over *u* converges to a generalized Pareto distribution as the threshold value *u* increases.

The distribution of X itself can be any of the commonly used distributions: normal, lognormal, *t*, etc., and will usually be unknown. The distribution of excess losses requires the estimation of two parameters, a positive scale parameter β and a shape or tail index parameter ξ. One must choose a threshold *u* that is high enough so that the theory applies but also low enough so that there are observations in excess of *u*.

Reference: Kevin Dowd, *Measuring Market Risk, 2nd Edition* (Wiley, 2005), Chapter 7 — Parametric Approaches (II): Extreme Value, pp. 201-202.
AIM: Describe the peaks over threshold (POT) approach.
Section: Market Risk Management and Measurement

5. A risk analyst is comparing the use of parametric and non-parametric approaches for calculating VaR and is concerned about some of the characteristics present in the loss data. Which of the following distribution characteristics would make parametric approaches the favored method to use?

- **a.** Skewness in the distribution
- **b.** Fat tails in the distribution
- **c.** Scarcity of high magnitude loss events
- **d.** Heteroskedasticity in the distribution

Correct answer: c

Explanation: Non-parametric approaches can accommodate fat tails, skewness, and any other non-normal features that can cause problems for parametric approaches. However, if the data period that is used in estimation includes few losses or losses with low magnitude, non-parametric methods will often produce risk measures that are too low. Hence parametric methods would be more appropriate in those situations.

Reference: Kevin Dowd, *Measuring Market Risk, 2nd Edition* (West Sussex, England: John Wiley & Sons, 2005), Chapter 4.
AIM: Describe the advantages and disadvantages of non-parametric estimation methods.
Section: Market Risk Management and Measurement

6. Lin Ping is valuing a 1-year credit default swap (CDS) contract which will pay the buyer 75% of the face value of a bond issued by Xiao Corp. immediately after a default by Xiao. To purchase this CDS, the buyer will pay the CDS spread, which is a percentage of the face value, once at the end of the year. Lin estimates that the risk-neutral default probability for Xiao is 5% per year. The risk-free rate is 3% per year. Assuming defaults can only occur halfway through the year and that the accrued premium is paid immediately after a default, what is the estimate for the CDS spread?

- **a.** 380 basis points
- **b.** 385 basis points
- **c.** 390 basis points
- **d.** 400 basis points

Correct answer: c

Explanation: The key to CDS valuation is to equate the present value (PV) of payments to the PV of expected payoff in the event of default. Let s denote the CDS spread.
π = probability of default during year 1 = 5%
C = contingent payment in case of default = 75%
d_i = discount factor = $e^{-0.03 \times 1}$ for 1-year and $e^{-0.03 \times 0.5}$ for half a year = 0.97044 and 0.98511
s = CDS spread (to be solved)

The premium leg, which includes the spread payment and accrual, is:

$s^*(0.5d_{0.5}{}^* \pi + d_1 (1-\pi)$) = s*(0.02463+0.92192) = s*0.94655

The payoff leg is:

$C * (d_{0.5}) * \pi$ = 0.03694

Solving for the spread: s*0.94655 = 0.03694 → s = 0.03902 or a spread of 390 basis points.

Reference: Allan Malz, *Financial Risk Management: Models, History, and Institutions* (Hoboken, NJ: John Wiley & Sons, 2011), chapter 7, pp. 250-253.
AIM: Define the different ways of representing spreads. Compare and differentiate between the different spread conventions and compute one spread given others when possible.
Section: Credit Risk Measurement and Management

7. You are the risk manager at Vision, a small fixed-income hedge fund that specializes in bank debt. Vision's strategy utilizes both relative value and long-only trades using credit default swaps (CDS) and bonds. One of the new traders has the positions described in the table below.

Bank	Position	Credit Rating
SBU	Long USD 10 million CDS	A
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Some of Vision's newest clients are restricted from withdrawing their funds for three years. You are currently evaluating the impact of various default scenarios to estimate future asset liquidity. You have estimated that the marginal probability of default of the Stanos bond is 5% in Year 1, 10% in Year 2, and 15% in Year 3. What is the probability that the bond makes coupon payments for 3 years and then defaults at the end of Year 3?

a. 13%
b. 15%
c. 27%
d. 73%

Correct answer: a

Explanation: The probability that the bond defaults in year 3 can be modeled as a Bernoulli trial given by the following equation, where MP stands for marginal probability:

$$P\ (\text{Default at end of year 3}) = (1-MP_{\text{year 1 default}}) * (1 - MP_{\text{year 2 default}}) * MP_{\text{year 3 default}}$$

$$= (1- 0.05) * (1 - 0.10) * 0.15 = 0.1283 \text{ or } 12.83\%.$$

References: Allan Malz, *Financial Risk Management: Models, History, and Institutions* (Hoboken, NJ: John Wiley & Sons, 2011), chapter 7, p. 236.
John Hull, *Options, Futures and Other Derivatives, 8th Edition* (New York: Pearson Prentice Hall, 2012), chapter 23.
AIM: Explain how default risk for a single company can be modeled as a Bernoulli trial.
Section: Credit Risk Measurement and Management

8. Consider a 1-year maturity zero-coupon bond with a face value of USD 1,000,000 and a 0% recovery rate issued by Company A. The bond is currently trading at 80% of face value. Assuming the excess spread only captures credit risk and that the risk-free rate is 5% per annum, the risk-neutral 1-year probability of default on Company A is closest to which of the following?

- **a.** 2%
- **b.** 14%
- **c.** 16%
- **d.** 20%

Correct answer: c

Explanation: This can be calculated by using the formula which equates the future value of a risky bond with yield (y) and default probability (π) to a risk free asset with yield (r):

$$1 + r = (1 - \pi) * (1 + y) + \pi R$$

π = Probability of default; R = Recovery rate

In the situation where the recovery rate is assumed to be zero, the risk-neutral probability of default can be derived from the following equation:

$$1 + r = (1 - \pi) * (1 + y) - (1 - \pi) * (FV/MV)$$

where MV = market value and FV = face value.

Inputting the data into this equation yields $\pi = 1 - (800{,}000*1.05)/1{,}000{,}000$ = 0.16.

Reference: Allan Malz, *Financial Risk Management: Models, History, and Institutions* (Hoboken, NJ: John Wiley & Sons, 2011), chapter 6, p. 203.
AIM: Explain the relationship between the yield spread and the probability of default and calculate default probability of a debt security using the credit spread.
Section: Credit Risk Management and Measurement

9. Portland General Electric (PGE) was an Enron subsidiary that was able to survive after the Enron implosion. At that time, there was a trend towards electric utility downgrades, particularly for those utilities operating within larger corporate structures. PGE survived in part due to ring-fencing. Which of the following statements about ring fencing is correct?

- **a.** A ring-fencing assets approach is typically only useful when a low quality firm wants to finance a high-quality project.
- **b.** When ring-fencing assets, options for credit enhancement include overcollateralization and financial guarantees provided by the parent against default of the subsidiary.
- **c.** A subsidiary holding the ring-fenced assets may be able to gain a higher credit rating than the parent, allowing it to issue bonds on the assets at a lower cost.
- **d.** Because the parent does not retain an equity interest in the subsidiary holding the ring-fenced assets, the subsidiary is not consolidated on the parent's balance sheet.

Correct answer: c

Explanation: Ring fencing is often undertaken to provide a higher credit rating to a subsidiary than is available to the parent. Derivative product companies or unregulated subsidiaries of investment banks are examples of this structure. There are other reasons for ring fencing assets, including freeing the assets from restrictions, taxes or other laws specific to a particular country.

Ring-fencing can be useful in two main situations: either when a low-quality firm cannot finance a high-quality project, or when a high-quality firm does not want to run the risk of being the sole financier of a low-quality project. The parent cannot guarantee the ring fenced assets, as this would allow creditors of the subsidiary to seek relief through the parent in the event of default of the subsidiary. The purpose of ring fencing assets is to create a structure that is bankruptcy remote from the parent. The retention of equity is a common feature of ring fencing. A subsidiary may remain consolidated on the parent company's balance sheet in cases where the parent retains a substantial equity interest.

Reference: Christopher Culp, *The Structuring Process, Structured Finance and Insurance: The Art of Managing Capital and Risk* (Hoboken, NJ: John Wiley & Sons, 2006), Chapter 13 pp. 274-279.
AIM: Describe the process and benefits of ring-fencing assets.
Section: Credit Risk Measurement and Management

10. Bank A, a large international bank, engages in trading with counterparties throughout the world. Recently, it has started to pay more attention to wrong-way risk in its trading book. Which one of the following four scenarios would serve as an example of wrong-way risk from Bank A's perspective?

- **a.** Bank A has a large exposure to Bank B's equity, and Bank B offers to sell put options with long maturities on its own equity to Bank A.
- **b.** Bank A enters into a medium-term repurchase agreement with Bank B using several different types of debt issued by bank B as collateral.
- **c.** Bank A actively manages its credit portfolio using credit default swaps, and decides to sell long-term credit protection to Bank B.
- **d.** Bank A enters into a forward rate agreement with Bank B to deliver at LIBOR+2.5%.

Correct answer: a

Explanation: According to Section 101 of Basel III, "a bank is exposed to "specific wrong-way risk" if future exposure to a specific counterparty is highly correlated with the counterparty's probability of default. For example, a company writing put options on its own stock creates wrong-way exposures for the buyer that is specific to the counterparty."

Reference: Basel Committee on Banking Supervision (February 16, 2012), *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems* (December 2010, revised June 2011) Note: www.bis.org/publ/bcbs189.pdf.
AIM: Describe changes to the regulatory capital framework, including changes to: Risk coverage, the use of stress tests, the treatment of counter-party risk with credit valuations adjustments, the use of external ratings, and the use of leverage ratios.
Section: Operational and Integrated Risk Management

11. The Chief Risk Officer of your bank has put you in charge of operational risk management. As a first step, you collect internal data to estimate the frequency and severity of operational-risk-related losses. The table below summarizes your findings:

Frequency Distribution		Severity Distribution	
Number of Occurrences	**Probability**	**Loss (USD)**	**Probability**
0	0.6	1,000	0.5
1	0.3	100,000	0.4
2	0.1	1,000,000	0.1

Based on this information, what is your estimate of the expected loss due to operational risk?

- **a.** USD 20,000
- **b.** USD 70,250
- **c.** USD 130,600
- **d.** USD 140,500

Correct answer: b

Explanation: The expected loss can be calculated by multiplying the expected frequency and the expected severity.

Expected frequency is equal to:

$$(0 * 0.6) + (1 * 0.3) + (2 * 0.1) = 0.5,$$

Expected severity is equal to:

$$(1000 * 0.5) + (100{,}000 * 0.4) + (1{,}000{,}000 * 0.1) = 140{,}500$$

The expected loss is therefore:

$$0.5 * 140{,}500 = 70{,}250$$

Reference: Mo Chaudhury, "A Review of the Key Issues in Operational Risk Capital Modeling", *The Journal of Operational Risk*, Volume 5/Number 3, Fall 2010: pp. 37-66.
AIM: Describe how a loss distribution is obtained from frequency and severity distributions.
Section: Operational and Integrated Risk Management

12. In its efforts to enhance its enterprise risk management function, Countryside Bank introduced a new decision-making process based on economic capital that involves assessing sources of risk across different business units and organizational levels. Which of the following statements regarding the correlations between these risks is correct?

- **a.** Correlations between the risks in the asset and liability sides of the balance sheet can be changed by management decisions.
- **b.** Generally, correlations between broad risk types such as credit, market, and operational risk are well understood and are easy to estimate at the individual firm level.
- **c.** Correlations between business units are only relevant in deciding total firm-wide economic capital levels and are not relevant for decisions at the individual business unit or project level.
- **d.** The introduction of correlations into firm-wide risk evaluation will result in a total VaR that, in general, is greater than or equal to the sum of individual business unit VaRs.

Correct answer: a

Explanation: Management has the ability to influence the correlations between these risks by changing the asset/liability mix, so management decision-making is indeed quite relevant.

Reference: Brian Nocco and René Stulz, "Enterprise Risk Management: Theory and Practice," *Journal of Applied Corporate Finance* 18, No. 4 (2006): pp. 8-20.
AIM: Describe the role of and issues with correlation in risk aggregation.
Section: Operational and Integrated Risk Management

13. In recent years, large dealer banks financed significant fractions of their assets using short-term, often overnight, repurchase (repo) agreements in which creditors held bank securities as collateral against default losses. The table below shows the quarter-end financing of four broker-dealer banks. All values are in USD billions:

	Bank A	Bank B	Bank C	Bank D
Financial instruments owned	823	629	723	382
Pledged as collateral	272	289	380	155

In the event that repo creditors become nervous about a bank's solvency, which bank is least vulnerable to a liquidity crisis?

a. Bank A
b. Bank B
c. Bank C
d. Bank D

Correct answer: a

Explanation: A liquidity crisis could materialize if repo creditors become nervous about a bank's solvency and choose not to renew their positions. If enough creditors choose not to renew, the bank could likely be unable to raise sufficient cash by other means on such short notice, thereby precipitating a crisis. However, this vulnerability is directly related to the proportion of assets a bank has pledged as collateral.

Bank A is least vulnerable since it has the least dependence on short-term repo financing (i.e. the lowest percentage of its assets out of the four banks is pledged as collateral:

272/823, or 33%

Reference: Darrell Duffie (2010), "Failure Mechanics of Dealer Banks", *Journal of Economic Perspectives* (24:1), pp. 51-72.
AIM: Identify factors that can precipitate or accelerate a liquidity crisis at a dealer bank and what prudent risk management steps can be taken to mitigate these risks.
Section: Operational and Integrated Risk Management

14. Galileo Vehicles (GV) and Leonardo Motors (LM) are both leading car manufacturers in hybrid car designs. Earlier this year, both companies introduced new hybrid models that are comparable to each other in almost every category. However, after both companies release pricing for their new models, LM's model is 20% less expensive than GV's. As a result, GV's stock price declined sharply while LM's stock price rose dramatically. Subsequently LM and GV announce that they have entered into merger discussions where the terms of the planned merger would give GV shareholders 1 share of LM per 3 shares of GV previously held. Post the announcement, GV's stock is trading at USD 20 and LM's stock is trading at USD 58. If you are confident that the merger will be completed, assuming zero transaction costs, which of the following investments should you make?

- **a.** Buy 300 shares of GV and short 100 shares of LM.
- **b.** Short 300 shares of GV and buy 100 shares of LM.
- **c.** Buy 300 shares of GV and buy 100 shares of LM.
- **d.** Short 300 shares of GV and short 100 shares of LM.

Correct answer: b

Explanation: If the merger goes through, the companies' prices should correspond on a 3:1 basis, with 1 share of LM corresponding to 3 shares of GV. However, at the given trading prices the ratio does not hold, with one share of LM being equal to USD 58 / USD 20, or 2.9 shares of GV. This shows that LM is undervalued compared to GV given the terms of the merger agreement. If the merger is completed, LM's stock will appreciate and/or GV's stock will depreciate relative to each other until the ratio reaches 3:1.

In order to exploit this potential arbitrage opportunity, you can short 300 shares of the relatively overvalued stock GV, resulting in a cash inflow of USD 6000, while buying 100 shares of the relatively undervalued stock LM for USD 5800, resulting in a net cash inflow of USD 200. If the merger is completed, then the long and the short positions will exactly offset each other given the 3:1 ratio and the trade will be closed. The original cash inflow of USD 200 would be your profit from this arbitrage trade if the merger is completed.

Reference: David P. Stowell (2010), *An Introduction to Investment Banks, Hedge Funds, and Private Equity*, Chapter 12.
AIM: Describe and interpret a numerical example of the following strategies: merger arbitrage, pairs trading, distressed investing and global macro strategy.
Section: Risk Management and Investment Management

15. At the end of 2007, Chad & Co.'s pension had USD 350 million worth of assets that were fully invested in equities and USD 180 million in fixed-income liabilities with a modified duration of 14. In 2008, the widespread effects of the subprime crisis hit the pension fund, causing its investment in equities to lose 50% of their market value. In addition, the immediate response from the government — cutting interest rates — to salvage the situation, caused bond yields to decline by 2%. What was the change in the pension fund's surplus in 2008?

- **a.** USD -55.4 million
- **b.** USD -124.6 million
- **c.** USD -225.4 million
- **d.** USD -230.4 million

Correct answer: c

Explanation: The change in the pension fund's surplus for the year 2008 is equal to the initial surplus S_0 at the end of 2007 less the ending surplus S_1 at the end of 2008. The initial surplus is calculated as $S_0 = A_0 - L_0 = 350 - 180 = 170$, where A_0 = the firm's initial assets and L_0 the firm's initial liabilities.

Next we have to calculate the surplus at the end of 2008. Given the 50% decline in the equity market, the new level of assets A_1 at the end of 2008 is equal to:

$(1 - 0.5) * 350$, or 175

The new level of liabilities L_1 can be calculated as:

$L_1 = (1 - (MD * \Delta y)) * L_0$

where MD is the modified duration, and

Δy is the change in yield.

Liabilities at end of 2008 are equal to:

$L_1 = (1 - (14 * -0.02)) * 180 = 230.4$.

Therefore the 2008 surplus S_1 is equal to $A_1 - L_1 = 175 - 230.4 = -55.4$ (which implies the pension fund is actually in a deficit situation at the end of 2008). The change in surplus for 2008 is hence $S_1 - S_0 = -55.4 - 170 = -225.4$ million.

Reference: Philippe Jorion, *Value-at-Risk: The New Benchmark for Managing Financial Risk*, *3rd Edition* (New York: McGraw-Hill, 2007), Chapter 17 — VaR and Risk Budgeting in Investment Management, p. 433.
AIM: Describe the investment process of large investors such as pension funds.
Section: Risk Management and Investment Management

16. As investors found out that highly-rated securities backed by subprime mortgages were not risk-free, the subprime crisis affected other asset classes. Which of the following mechanisms played an important role in the transmission of the crisis from the subprime sector to other asset classes?

- **a.** Impact of cross-default clauses linking industrial bonds and commercial mortgages held by banks
- **b.** Increase in repo haircuts causing banks to sell off assets to meet collateral calls
- **c.** Tightening of discount window lending standards by the U.S. Federal Reserve
- **d.** Exercise of credit default swap contracts tied to subprime mortgage pools held in off-balance sheet vehicles

Correct answer: b

Explanation: A brief excerpt from the article provides a summary: "Uncertain about the solvency of counterparties, repo depositors became concerned that the collateral bonds might not be liquid; if all firms wanted to hold cash — a flight to quality — then collateral would have to decline in price to find buyers. This is the crucial link between the subprime shock and other asset categories".

This decline in the value of collateral became evident in the rapid increase in the "repo haircut", which is the percentage difference between the market value of the pledged collateral and the amount of funds lent. For example, with a 5% "repo haircut", a bank would only allow a USD 95 deposit against USD 100 face value of collateral. This repo haircut rose dramatically in late 2007 and 2008, from zero at the beginning of August 2007 to almost 10% in early 2008 and reaching over 45% by early 2009. As the value of collateral began to fall, this led to a deterioration in valuations across asset classes which also commenced in August 2007 when the LIBOR-OIS spread jumped. This occurred even though the subprime fundamentals (as measured by the ABX index of credit derivatives) had been deteriorating for months prior to that.

Reference: Gary Gorton (05-2009), *Slapped in the Face by the Invisible Hand: Banking and the Panic of 2007*, p. 33.
AIMS: Explain the function of and define repos, and discuss their use as the primary mechanism driving shadow banking. Explain how the shock from the subprime mortgage collapse affected asset classes that were unrelated and evolved into the 2007 banking system panic.
Section: Current Issues in Financial Markets

17. In the years leading up to the collapse of the Icelandic banking system, how did the relationship between the Central Bank of Iceland (CBI) and the Icelandic banks change?

- **a.** The CBI supplemented credit ratings from the major rating agencies with market-implied ratings when determining the liquidity to provide to the banks.
- **b.** The CBI began to issue loans to the banks that were denominated in Euros.
- **c.** The CBI began to issue more loans to the banks that were collateralized by the bonds of other Icelandic banks.
- **d.** The CBI began to issue loans to the banks that were denominated in U.S. dollars.

Correct answer: c

Explanation: In 2007 and 2008, the amount of Icelandic banks' debt held by other Icelandic banks grew dramatically, from around one percent of GDP in January 2007 to almost 30% of GDP by September 2008. In turn, a large proportion of these bonds was used as collateral at the CBI. One example occurred in 2008 when the Icelandic banks issued debt to the savings bank, Icebank, which then used the debt as collateral at the CBI. Other banks such as Kaupthing and Glitnir also issued covered bonds which were used for collateralized loans at the CBI.

CBI's rules for the credit standards of eligible collateral were broadly similar to the rules of the European System of Central Banks (ESCB), stating that unsecured bonds and bills were required to have a minimum long-term rating of "A-" by S&P or Fitch or "A3" by Moody's, as well as having their securities traded on a regulated market in the EEA. There was no use of market-implied ratings. The CBI only issued loans to the banks that were denominated in krona.

Reference: Arthur M. Berd (editor), *Lessons From the Financial Crisis* (London: Risk Books, 2010), Chapter 4: The Collapse of the Icelandic Banking System, pp. 111.
AIMS: Describe the severity of the banking crisis, the currency crisis, and the public debt crisis. Understand how banks funded their risky business models before and after the 2006 mini-crisis.
Section: Current Issues in Financial Markets

18. A portfolio has USD 2 million invested in Stock A and USD 1 million invested in Stock B. The 95% 1-day VaR for each individual position is USD 40,000. The correlation between the returns of Stock A and Stock B is 0.5. While rebalancing, the portfolio manager decides to sell USD 1 million of Stock A to buy USD 1 million of Stock B. Assuming that returns are normally distributed and that the rebalancing does not affect the volatility of the individual stocks, what effect will this have on the 95% 1-day portfolio VaR?

- **a.** There will be no effect.
- **b.** It will increase by USD 20,370.
- **c.** It will increase by USD 21,370.
- **d.** It will increase by USD 22,370.

Correct answer: d

Explanation: The first step is to calculate the VaR of the original portfolio. This can be done by using the following equation:

$$VaR_{Port\ (A,B)} = \sqrt{(VaR_A^2 + VaR_B^2 + (2\rho * VaR_A * VaR_B)}$$

where ρ is the correlation coefficient.

Portfolio VaR (before):

$$\sqrt{40000^2 + 40000^2 + (2 * 0.5 * 40000 * 40000)} = \text{USD } 69{,}282.$$

After the rebalance, the market value of the position in Stock A is halved, so VaR(A) is now equal to $20,000. Meanwhile the market value for the position in B has doubled so that VaR(B) is now $80,000. Hence we can now calculate the VaR of the new portfolio as follows:

Portfolio VaR (after) = $\sqrt{20000^2 + 80000^2 + (2 * 0.5 * 20000 * 80000)}$ = USD 91,652.

So the VaR will increase by (91,652 – 69,282), or USD 22,370.

Reference: Philippe Jorion, *Value-at-Risk: The New Benchmark for Managing Financial Risk, 3rd Edition*, Chapter 7: Portfolio Risk — Analytical Methods, pp. 161-164.
AIM: Compute diversified VaR, individual VaR, and undiversified VaR of a portfolio.
Section: Risk Management and Investment Management

19. In calculating its risk-adjusted return on capital, your bank uses a capital charge of 2.50% for revolving credit facilities with a loan equivalent factor of 0.35 assigned to the undrawn portion. Recently, you have become concerned that the protective covenants embedded in these loans are weak and may not prevent customers from drawing on the facilities during times of stress. As such, you have recommended doubling the loan equivalent factor to 0.70. This recommendation has met with resistance from the loan origination team, and senior management has asked you to quantify the impact of your recommendation. For a typical facility that has an original principal of USD 1 billion and is 30% drawn, how much additional economic capital would have to be allocated if you increase the loan equivalent factor from 0.35 to 0.70?

- **a.** USD 3.50 million
- **b.** USD 6.13 million
- **c.** USD 8.75 million
- **d.** USD 13.63 million

Correct answer: b

Explanation: The required economic capital to support a loan in the RAROC model can be calculated using the following formula:

$$\textit{Required Capital} = [B_{DRAWN} + (B_{UNDRAWN} * \textit{LEF})] * \textit{CF}$$

where LEF represents the loan equivalent factor and CF represents the capital factor.

Therefore the initial required economic capital is calculated as follows:
[(1 billion * 0.3) + (1 billion * 0.7 * 0.35)] * 2.5% = USD 13.625 million,

and the required capital if the change is implemented would be:
[(1 billion * 0.3) + (1 billion * 0.7 * 0.70)] * 2.5% = USD 19.75 million.

Hence the additional required economic capital would be 19.75 – 13.625 or 6.13 million.

Reference: Michel Crouhy, Dan Galai and Robert Mark, *Risk Management* (New York: McGraw-Hill, 2001), Chapter 14, p. 550.
AIM: Compute and interpret the RAROC for a loan or loan portfolio, and use RAROC to compare business unit performance.
Section: Operational and Integrated Risk Management

20. Which of the following statements regarding frictions in the securitization of subprime mortgages is correct?

- **a.** The arranger will typically have an information advantage over the originator with regard to the quality of the loans securitized.
- **b.** The originator will typically have an information advantage over the arranger, which can create an incentive for the originator to collaborate with the borrower in filing false loan applications.
- **c.** The major credit rating agencies are paid by investors for their rating service of mortgage-backed securities, and this creates a potential conflict of interest.
- **d.** The use of escrow accounts for insurance and tax payments eliminates the risk of foreclosure.

Correct answer: b

Explanation: One of the key frictions in the process of securitization involves an information problem between the originator and arranger. In particular, the originator has an information advantage over the arranger with regard to the quality of the borrower. Without adequate safeguards in place, an originator can have the incentive to collaborate with a borrower in order to make significant misrepresentations on the loan application. Depending on the situation, this could be either construed as predatory lending (where the lender convinces the borrower to borrow too large of a sum given the borrower's financial situation) or predatory borrowing (the borrower convinces the lender to lend too large a sum).

The major rating agencies are not paid by the investors. Escrow accounts can forestall but not eliminate the risk of foreclosure.

Reference: Adam Ashcroft and and Til Schuermann, "Understanding the Securitization of Subprime Mortgage Credit", FRB of NY Staff reports, No. 318 (March 2008), page i.
AIM: Identify and describe key frictions in subprime mortgage securitization, and assess the relative contribution of each factor to the subprime mortgage problems.
Section: Credit Risk Measurement and Management

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